Complex Actions, Sign Problems, and Contour Deformations in QFT

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Abstract

We survey the sign problem in Euclidean lattice QFT and the contour deformation methods used to address it, with emphasis on the Lefschetz thimble decomposition and its application to the O(N) model at large N. We connect the steepest descent contour strategy of `mass-gap-v3` to the broader thimble literature, identify the key mathematical steps needed for a rigorous proof, and discuss both positive results and fundamental obstructions.

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1.1 General setup

A lattice QFT partition function has the form

$$Z = \int_{\mathbb{R}^n} e^{-S(\phi)}\, d\phi, \tag{1}$$

where $\phi \in \mathbb{R}^n$ is the field configuration ($n = N|\Lambda|$ for N-component fields on lattice Λ) and $S : \mathbb{R}^n \to \mathbb{R}$ is the Euclidean action. When S is real, $e^{-S} > 0$ and the integrand defines a probability measure (after normalization). Observables are expectations under this measure.

The **sign problem** arises when a change of variables or auxiliary field introduces a *complex* action $S : \mathbb{R}^n \to \mathbb{C}$, so e^{-S} is complex-valued. The integrand is no longer a positive measure, and Monte Carlo methods fail because importance sampling requires positive weights.

1.2 Sources of complex actions

Complex actions arise in several settings:

- **Finite density**: Chemical potential μ gives $\det(D + \mu\gamma_0)$ which is complex for QCD.
- **Real-time evolution**: The Minkowski action iS_M is purely imaginary.
- **Hubbard-Stratonovich transformation**: The HS identity for the Euclidean quartic $e^{-\lambda a^2}$ requires imaginary coupling $e^{i\sigma a}$ (Section 2).
- **Topological terms**: The θ-term $e^{i\theta Q}$ is complex for $\theta \neq 0$.
- **Fermion determinant**: $\det(D)$ can be complex for chiral or finite-density theories.

1.3 Why contour deformation?

Since $S(\phi)$ extends to a holomorphic (or meromorphic) function $S : \mathbb{C}^n \to \mathbb{C}$, the integral (1) can be deformed from $\mathbb{R}^n$ to any n-dimensional cycle $\Gamma \subset \mathbb{C}^n$ in the same homology class, provided Γ avoids singularities and the integrand decays at infinity:

$$Z = \int_\Gamma e^{-S(\sigma)}\, d^n\sigma. \tag{2}$$

The goal is to choose Γ so that e^{-S} is as close to positive (non-oscillatory) as possible on Γ.

2 The HS sign problem

2.1 The Euclidean quartic requires imaginary coupling

For the $\mathrm{O}(N)$ LSM, the quartic interaction enters as $e^{-\lambda a^2}$ with $a = |\varphi|^2/N - v^2$. The HS identity:

$$e^{-\lambda a^2} = \frac{1}{\sqrt{4\pi\lambda}} \int_{-\infty}^{\infty} e^{-\sigma^2/(4\lambda) + i\sigma a}\, d\sigma. \tag{3}$$

The coupling $i\sigma a$ is **imaginary** because $e^{-\lambda a^2}$ is a Gaussian in a (not $e^{+\lambda a^2}$). This is forced by Euclidean signature.

2.2 The σ-effective action

After applying HS at each site and integrating out φ:

$$Z = c\int_{\mathbb{R}^{|\Lambda|}} e^{f(\sigma)}\, d\sigma, \qquad f(\sigma) = -\sum_x \frac{\sigma(x)^2}{4\lambda} - \frac{N}{2}\mathrm{Tr}\log(-\Delta + 2i\sigma z) - iv^2 \sum_x \sigma(x), \tag{4}$$

where $z = 1/\sqrt{N}$. This integral converges on $\mathbb{R}^{|\Lambda|}$ (the Gaussian factor provides decay, and $|\det(-\Delta + 2i\sigma z)^{-1}| \le \det(-\Delta)^{-1}$), but the integrand is **complex**.

2.3 Comparison: NLSM vs LSM

For the NLSM (nonlinear sigma model with constraint $|\varphi|^2 = N\beta$), a real mass m_0^2 can be added to the GFF for free (it is constant on the constraint surface). The operator $-\Delta + m_0^2 - 2i\alpha z$ has positive real part, and the FK bound gives single-σ decay $|G_\sigma(x,0)| \le G_{m_0}(x,0) \sim e^{-m_0|x|}$.

For the LSM: the add-subtract trick cancels the real mass, leaving $-\Delta - 2i\sigma$ with **no real part** beyond the kinetic term. The mass gap comes entirely from the oscillatory σ-average. See `mass-gap-v3.tex`, Sections 3–4.

3 Lefschetz thimbles

3.1 Morse theory for complex integrals

Consider a holomorphic function $f : \mathbb{C}^n \to \mathbb{C}$ (the exponent in the partition function). The critical points of f are the solutions of $\partial f/\partial\sigma_i = 0$. For each critical point σ_*, the **Lefschetz thimble** $\mathcal{J}_{\sigma_*}$ is the union of gradient flow lines of $\mathrm{Re}\, f$ that flow *into* σ_* as $t \to +\infty$:

$$\mathcal{J}_{\sigma_*} = \left\{ \sigma(0) : \frac{d\sigma}{dt} = \overline{\nabla f(\sigma)},\ \lim_{t\to+\infty} \sigma(t) = \sigma_* \right\}. \tag{5}$$

Key properties of $\mathcal{J}_{\sigma_*}$:

1. $\mathcal{J}_{\sigma_*}$ is an n-dimensional (real) submanifold of $\mathbb{C}^n$.
2. $\mathrm{Im}\, f = \mathrm{Im}\, f(\sigma_*)$ is **constant** on $\mathcal{J}_{\sigma_*}$ (the flow preserves the imaginary part).
3. $\mathrm{Re}\, f$ is **maximized** at σ_* on $\mathcal{J}_{\sigma_*}$ (steepest descent from the saddle).
4. The integrand $e^{f(\sigma)}$ is proportional to a **real positive** function times the constant phase $e^{i\,\mathrm{Im} f(\sigma_*)}$ on $\mathcal{J}_{\sigma_*}$.

3.2 Thimble decomposition

The original integration cycle $\mathbb{R}^n$ can be decomposed:

$$[\mathbb{R}^n] = \sum_{\sigma_*} n_{\sigma_*}\, [\mathcal{J}_{\sigma_*}], \tag{6}$$

where $n_{\sigma_*} \in \mathbb{Z}$ are **intersection numbers** (the intersection of $\mathbb{R}^n$ with the *dual* thimble $\mathcal{K}_{\sigma_*}$, defined by the ascending gradient flow). Therefore:

$$Z = \sum_{\sigma_*} n_{\sigma_*} \int_{\mathcal{J}_{\sigma_*}} e^{f(\sigma)}\, d\sigma = \sum_{\sigma_*} n_{\sigma_*}\, e^{i\,\mathrm{Im} f(\sigma_*)}\, Z_{\sigma_*}, \tag{7}$$

where $Z_{\sigma_*} = \int_{\mathcal{J}_{\sigma_*}} e^{\mathrm{Re} f(\sigma) - \mathrm{Re} f(\sigma_*)}\, |d\sigma| > 0$ is **real and positive**.

3.3 When is the sign problem solved?

The sign problem is eliminated on a single thimble (since the integrand is proportional to a positive function). But the *inter-thimble* sign problem remains if multiple thimbles contribute with different phases $e^{i\,\mathrm{Im} f(\sigma_*)}$.

The sign problem is fully solved when:

(a) Only one thimble has $n_{\sigma_*} \neq 0$, or

(b) All contributing thimbles have the same phase $\mathrm{Im} f(\sigma_*)$, or

(c) One thimble dominates exponentially: $\mathrm{Re} f(\sigma_*) \gg \mathrm{Re} f(\sigma'_*)$ for all other contributing critical points σ'_*.

3.4 Stokes phenomenon

As parameters (coupling constants, volume) vary, the thimble decomposition (6) can jump: when two critical points have $\mathrm{Im} f(\sigma_*) = \mathrm{Im} f(\sigma'_*)$, a gradient flow line connects them, and the intersection numbers change. These **Stokes walls** are codimension-1 surfaces in parameter space.

Cao-Santachiara-Usciati [4] found infinitely many Stokes walls for lattice Liouville theory as $c \to (-\infty, 1]$, showing the thimble structure can be very intricate.

4 Contour deformation methods in lattice QFT

4.1 Exact thimble methods

The pioneering work of Cristoforetti-Scorzato et al. proposed Monte Carlo sampling directly on a Lefschetz thimble. The main challenge: computing the **residual phase** (the Jacobian of the map from $\mathbb{R}^n$ to $\mathcal{J}_{\sigma_*}$), which is expensive ($O(n^3)$ per configuration).

4.2 Generalized thimble / gradient flow

Instead of the exact thimble, one can use the **flowed manifold**:

$$\mathcal{M}_T = \left\{\sigma(T) : \sigma(0) \in \mathbb{R}^n,\ \dot{\sigma} = \overline{\nabla f(\sigma)}\right\}, \tag{8}$$

obtained by flowing the real domain along the anti-holomorphic gradient of f for time T. As $T \to \infty$: $\mathcal{M}_T \to \bigcup_{\sigma_*} \mathcal{J}_{\sigma_*}$. For finite T: $\mathcal{M}_T$ is a smooth deformation of $\mathbb{R}^n$ that already reduces the sign problem.

Advantage: No need to identify critical points or compute intersection numbers. The flow is a well-posed ODE.

4.3 Worldvolume HMC (Fukuma et al.)

Fukuma and collaborators [5, 6, 7, 9] developed the most systematic computational approach:

1. Foliate $\mathbb{C}^n$ by a continuum family of integration surfaces $\{\mathcal{M}_T\}_{T \geq 0}$.
2. Introduce a symplectic structure on the tangent bundle of each $\mathcal{M}_T$.
3. Perform HMC (Hamiltonian Monte Carlo) on $\mathcal{M}_T$, treating the surface as a Riemannian manifold.

No Jacobian computation is needed for configuration generation (only for reweighting). Extended to group manifolds (gauge theories) in [9]. Tested on the Hubbard model [8] with severe sign problem.

4.4 Constant contour shifts

The simplest contour deformation: $\sigma \to \sigma + i\delta$ (constant imaginary shift). Ostmeyer et al. [16] proved this completely eliminates the sign problem in the Hubbard model at large chemical potential. Gaentgen et al. [10] applied this to C_{20} and C_{60} fullerenes.

4.5 Fundamental limitations

Lawrence-Yamauchi [15] proved that for 2D Abelian lattice gauge theory, the sign problem is **exponential** and **cannot be removed by any contour deformation**. The obstruction is topological: the compact gauge group $U(1)$ has $\pi_1(U(1)) = \mathbb{Z}$, creating nontrivial winding sectors that contribute with different phases.

Remark 1. This negative result does **not** apply to scalar field theories like the O(N) model. The field space $\mathbb{R}^n$ is contractible ($\pi_k(\mathbb{R}^n) = 0$ for all k), so there is no topological obstruction to contour deformation. The question for scalar theories is analytic (singularity avoidance, decay at infinity), not topological.

5 Application to the O(N) model at large N

5.1 The exponent and its critical points

The exponent (4) extends to a holomorphic function $f : \mathbb{C}^{|\Lambda|} \to \mathbb{C}$ (away from the singularities of the log det). The critical point equation $\partial f/\partial\sigma(x) = 0$ gives:

$$\frac{\sigma_*(x)}{2\lambda} = -iNz\,(-\Delta + 2i\sigma_* z)^{-1}(x,x) - iv^2. \tag{9}$$

For the translation-invariant saddle $\sigma_* = \sigma_0$ (constant), this becomes the **gap equation**:

$$\frac{\sigma_0}{2\lambda} = -iz\big[N\,C_{\sigma_0}(x,x) + v^2\big], \qquad C_{\sigma_0} = (-\Delta + 2i\sigma_0 z)^{-1}. \tag{10}$$

At $\sigma_0 = 0$ this reads $0 = -iz[NC_0(0,0) + v^2]$, which determines v^2 (or equivalently, the bare coupling).

5.2 Single-thimble dominance at large N

The exponent scales as $f \sim N \cdot g(\sigma)$ for a function g independent of N. Therefore:

- The Hessian scales as $f'' \sim N$, giving saddle-point width $\sim 1/\sqrt{N}$.
- At competing saddle points σ'_*: the action difference $\mathrm{Re} f(\sigma_*) - \mathrm{Re} f(\sigma'_*) \sim N \cdot \Delta g > 0$ grows linearly in N.
- By condition (c) of Section 3.3: the dominant thimble contribution exceeds all others by a factor e^{cN} for some $c > 0$.

Conclusion: For N sufficiently large, the single thimble $\mathcal{J}_{\sigma_*}$ through the dominant saddle captures Z up to exponentially small corrections:

$$Z = n_{\sigma_*}\, e^{i\,\mathrm{Im} f(\sigma_*)}\, Z_{\sigma_*}\big(1 + O(e^{-cN})\big). \tag{11}$$

5.3 Properties of the dominant thimble

On $\mathcal{J}_{\sigma_*}$:

1. The integrand $e^{f(\sigma)}$ is proportional to a **positive** function (Section 3.1, property 4).
2. Near σ_*: $f \approx f(\sigma_*) - \frac{1}{2}(\delta\sigma,\, H\,\delta\sigma)$ with $H = -f''(\sigma_*) > 0$ (positive definite Hessian). The measure is **log-concave** near the saddle.
3. The φ-coupling becomes effectively **real** on the thimble (since $\mathrm{Im} f = \text{const}$ constrains the imaginary parts of σ). This is needed for the FK bound.

5.4 The proof strategy

Combining the thimble structure with the Brascamp-Lieb and Feynman-Kac ingredients:

1. **Thimble existence**: Show $\mathcal{J}_{\sigma_*}$ is a smooth $|\Lambda|$-dimensional manifold avoiding the singularities $\det(-\Delta + 2i\sigma z) = 0$.
2. **Single-thimble dominance**: For $N > N_0$, the intersection number $n_{\sigma_*} = 1$ and all other thimbles are suppressed by e^{-cN}.
3. **Cauchy's theorem**: The deformation from $\mathbb{R}^{|\Lambda|}$ to $\mathcal{J}_{\sigma_*}$ is valid (no singularities crossed, boundary terms vanish).
4. **Brascamp-Lieb on the thimble**: The measure $e^{\mathrm{Re} f}|_{\mathcal{J}_{\sigma_*}}$ is log-concave (at least near the saddle). Apply BL for σ-concentration: $\mathrm{Var}(\sigma(x)) \leq 1/(\kappa N)$. Here Conlon-Dabkowski [2] extends BL to lattice field models with convex gradient action, which is the relevant setting.
5. **Real coupling on the thimble**: On $\mathcal{J}_{\sigma_*}$, the constraint $\mathrm{Im} f = \text{const}$ makes the φ-operator $-\Delta + V(\sigma)$ have a **real** potential V. The FK bound / Borell sub-Gaussian estimate gives exponential decay.
6. **Mass gap**: Combine BL concentration + FK bound to get $m > 0$ for $N > N_0$.

5.5 The flowed manifold alternative

Instead of the exact thimble, one can work with the flowed manifold $\mathcal{M}_T$ for large but finite T. Advantages:

- No need to identify all critical points.
- $\mathcal{M}_T$ is diffeomorphic to $\mathbb{R}^{|\Lambda|}$ (no topology change).
- The sign problem is reduced (not eliminated) but the residual oscillations are bounded by $e^{-c(T)N}$ for a function $c(T)$ increasing in T.

For a rigorous proof: take T large enough that the residual oscillations are smaller than the mass gap signal.

6 Related approaches and results

6.1 Complex Langevin

Complex Langevin (CL) dynamics extends stochastic quantization to complex actions by complexifying the field variables and evolving with a complex drift. CL avoids the sign problem by construction but can converge to **wrong** results.

Boguslavski-Hotzy-Muller [1] showed that CL converges correctly when there is a **single dominant compact thimble**. Multiple thimbles with different phases cause CL to fail. This connects CL to the thimble picture and suggests that single-thimble dominance (which holds at large N) implies CL correctness.

Cai-Dong-Kuang [3] gave the first rigorous mathematical analysis of CL, proving that localized probability densities do not exist for general compact groups, explaining CL instability for gauge theories.

6.2 Brascamp-Lieb extensions

Conlon-Dabkowski [2] extended BL to lattice field models in $d \geq 2$ with uniformly convex gradient action. They proved charge-charge correlations in the Coulomb dipole gas are close to Gaussian, going beyond Dimock-Hurd and Conlon-Spencer. The key observation: the sine-Gordon measure is invariant for a stochastic dynamics, allowing a dynamical proof of correlation decay.

This is directly relevant to the BL step on the thimble: the σ-measure on $\mathcal{J}_{\sigma_*}$ has a convex gradient action near the saddle, and Conlon-Dabkowski's extension handles precisely this setting.

6.3 $\mathrm{P}(\varphi)_2$ via Segal axioms

Lin [14] constructed the $\mathrm{P}(\varphi)_2$ Gibbs state on infinite-volume periodic surfaces and proved the mass gap using Segal's axioms and a "Bayes principle" for conditional probabilities. This is the first construction of an interacting QFT on surfaces of infinite genus with a mass gap.

This provides an alternative approach to the $N = 1$ test case ($\mathrm{P}(\varphi)_2$ mass gap), complementing the HS/thimble strategy and the classical Glimm-Jaffe-Spencer approach.

6.4 Constructive RG

Giuliani-Mastropietro-Rychkov-Scola [11] continued the rigorous constructive RG program, constructing scale-invariant response functions at a non-trivial fixed point for fermionic ψ_d^4 (using convergent tree expansion). While the model differs from ours, the convergent expansion techniques are potentially applicable to the $1/N$ expansion.

7 Open questions

1. **Singularity avoidance**: Can we prove the dominant thimble $\mathcal{J}_{\sigma_*}$ avoids $\det(-\Delta + 2i\sigma z) = 0$? The zeros are at σ such that $-2iz\sigma$ is an eigenvalue of $-\Delta$, i.e., $\sigma = ik^2/(2z)$

(purely imaginary, distance $\geq m_0^2/(2z)$ from the real saddle). For large N ($z = 1/\sqrt{N}$ small), these are far from the saddle.

2. **Global log-concavity**: Is $\mathrm{Re}f$ concave on *all* of $\mathcal{J}_{\sigma_*}$, or only near the saddle? The $\mathrm{Tr}\log$ term introduces non-convexity far from the saddle. For BL: we need convexity at least within the $O(1/\sqrt{N})$-width peak, which holds by the Hessian computation.

3. **Intersection numbers**: Can we compute n_{σ_*} rigorously? For the dominant saddle on the real axis with f'' positive definite, homotopy arguments should give $n_{\sigma_*} = 1$.

4. **N=1 test case**: In $d = 0$ (one site), f is a function of one complex variable. The thimble is a curve in $\mathbb{C}$ that can be computed explicitly. Does the thimble analysis reproduce the known $\mathrm{P}(\varphi)_2$ mass gap?

5. **Comparison with Kupiainen**: Kupiainen's NLSM proof uses the constraint trick (real mass for free) and avoids contour deformation entirely. Can the thimble approach give a simpler proof of the NLSM mass gap? Or a proof of the **LSM** mass gap (which Kupiainen did not prove)?

6. **Threshold** N_0: What is the minimal N for single-thimble dominance? Kupiainen's threshold is $N > C\,m_0^{-25}$ (very large). Can the thimble approach improve this?

8 Summary

The Lefschetz thimble decomposition provides a natural framework for the steepest descent contour strategy of `mass-gap-v3`. The key facts:

- The "steepest descent contour" $\Gamma = \{\mathrm{Im}f = 0\}$ through the saddle is exactly a Lefschetz thimble.
- On each thimble, the integrand is proportional to a positive function — the sign problem is solved per-thimble.
- At large N, a single thimble dominates (saddle-point sharpness), so the inter-thimble sign problem is also controlled.
- The Lawrence-Yamauchi obstruction (for gauge theories) does not apply to scalar models: the field space $\mathbb{R}^n$ is contractible.
- On the thimble, the σ-measure is real, positive, and approximately log-concave — BL and FK bounds apply.
- The flowed manifold approach (Fukuma) offers a more robust alternative that avoids identifying exact thimbles.

The main mathematical challenges are: (1) proving singularity avoidance, (2) establishing log-concavity on the thimble, and (3) making the FK bound work with the effective real potential. These are the subjects of ongoing work in `mass-gap-v3` and the Lean formalization.

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